

EMCH380 – Project Management

Project Planning Lifecycle Phases

Slides are developed from multiple references.

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Project Management Objectives

Project Planning Lifecycle Phases

PROJECT MANAGEMENT

- Project Management involves managing the 2 resources:
 - Time
 - Money
- Major causes of poor project performance (Cost or Schedule overrun) are:
 - Bad Planning
 - Bad Scheduling

PROJECT PLANNING OBJECTIVES

- To eliminate or reduce uncertainty
- To improve efficiency of the project
- To provide a basis for monitoring and controlling work

PROJECT PLANNING LIFECYCLE PHASES (BY KERZNER)

- Conceptualization
- Feasibility
- Preliminary Planning
- Detailed Planning
- Execution
- Testing and Commissioning

Project Planning

Project Planning Lifecycle Phases

PROJECT PLANNING

- At the end of each phase of project planning there is a meeting between the project manager, sponsor, senior management, and even the customer to assess accomplishments of this life cycle phase and to get approval for the next phase
- These meetings are called critical design reviews or gates.

PROJECT PLANNING LIFECYCLE PHASES (BY KERZNER)

- **Conceptualization**
- Feasibility
- Preliminary Planning
- Detailed Planning
- Execution
- Testing and Commissioning

A. CONCEPTUALIZATION

- This phase includes brainstorming and common sense
- It involves two critical factors:
 - identify and define the problem
 - identify and define prudential solutions
- Brainstorming sessions works best if no formal authority is present
- Sessions should not exceed 60 minutes.

PROJECT PLANNING LIFECYCLE PHASES (BY KERZNER)

- Conceptualization
- **Feasibility**
- Preliminary Planning
- Detailed Planning
- Execution
- Testing and Commissioning

B. FEASIBILITY

- Feasibility study considers the technical aspects of the conceptual alternatives
- Experience & technical expertise are required to gather proper information & analyze it.
- Estimate the probable elapsing time, staffing and equipment requirements.

B. FEASIBILITY (CONT'D)

- Identify the probable costs & consequences of investing in the new project
- User involvement during the feasibility study is critical to judge impact of alternative approaches
- Provide management with the predictable results of implementing a specific project

B. FEASIBILITY (CONT'D)

- It provides a firmer basis on which to decide to undertake project or not
- End result is management decision on whether to terminate the project or to approve its next phase.

PROJECT PLANNING LIFECYCLE PHASES (BY KERZNER)

- Conceptualization
- Feasibility
- **Preliminary Planning**
- Detailed Planning
- Execution
- Testing and Commissioning

C. PRELIMINARY PLANNING

- In this phase, effort is officially defined as a project:
 - Identify and prioritize project objectives
 - Identify major elements of work required to satisfy objectives and how are they interrelated
 - Identify functional divisions that will assume responsibility for accomplishing objectives and major element work requirement
 - Identify if organizational resources are available
 - Identify information requirements for the project

C. PRELIMINARY PLANNING INFORMATION REQUIREMENTS

- The following core documents are considered:
 - Statement of work (SOW)
 - Project Specifications
 - Milestone Schedule

C. PRELIMINARY PLANNING – STATEMENT OF WORK

- Is a narrative description of the work required for the project.
It includes
 - Objectives of the project
 - Brief description of the work
 - Funding constraints if one exists
 - The specifications
 - A “gross” schedule including start and end date of project and major milestones and written reports (data items)

C. PRELIMINARY PLANNING – STATEMENT OF WORK (CONT'D)

- SOWs should be carefully written to avoid misinterpretations which in turn can result in dollar losses
- One reason for misinterpretations is using imprecise language
- What follows are examples of this.

C. PRELIMINARY PLANNING – WRITING SOW EXAMPLE

II

- Case 1:

The SOW says that you are to conduct a minimum of fifteen tests to determine the materials properties of a new substance.

C. PRELIMINARY PLANNING – WRITING SOW EXAMPLE (CONT'D)

II

- Solution 1:

You price out twenty substance just to “play it safe”.

C. PRELIMINARY PLANNING – WRITING SOW EXAMPLE (CONT'D)

II

- Problem 1:

At the end of the fifteenth test, the customer says that the results are inconclusive and that you must run another fifteen tests. The cost overrun is \$40,000.

C. PRELIMINARY PLANNING – WRITING SOW EXAMPLE (CONT'D)

II

- Case 2:

The Navy gives you a contract in which the SOW states that the prototype must be tested in “water”.

C. PRELIMINARY PLANNING – WRITING SOW EXAMPLE (CONT'D)

II

- Solution 2:

You drop the prototype into a swimming pool to test it.

C. PRELIMINARY PLANNING – WRITING SOW EXAMPLE (CONT'D)

II

- Problem 2:

Navy's definition of "water" is the Atlantic ocean, and it costs you \$1 million to transport all of your test engineers and test equipment to the Atlantic Ocean.

C. PRELIMINARY PLANNING – WRITING SOW EXAMPLE (CONT'D)

II

- Case 3:

You receive a contract in which the SOW says that you must transport goods across the country using “aerated” box cars.

C. PRELIMINARY PLANNING – WRITING SOW EXAMPLE (CONT'D)

- Solution 3:

You select boxcars that have open tops so that air can flow in.

C. PRELIMINARY PLANNING – WRITING SOW EXAMPLE (CONT'D)

II

- Problem 3:

During the trip, the train goes through an area of torrential rains, and the goods are ruined. The customer wanted boxcars that were aerated from below. The court is currently deciding who should be blamed for misinterpretation of the word “aerated”.

C. PRELIMINARY PLANNING – STATEMENT OF WORK (CONT'D)

- Complexity of SOW is determined by Top management &/ or user groups
- Misinterpretations of SOW can occur no matter how hard the quest for perfection is during the definition phase.
- Today both private industry and government agencies are developing manuals on SOW preparation

C. PRELIMINARY PLANNING – CHECKLIST FOR SOW PREPARATION

- Is the SOW specific enough to permit a contractor to make a tabulation and summary of manpower and resources needed to accomplish each SOW task element?
- Are specific duties of the contractor stated so he will know what is required, and can the contracting officer's representative, who signs the acceptance report, tell whether the contractor's has complied?

C. PRELIMINARY PLANNING – CHECKLIST FOR SOW PREPARATION (CONT'D)

II

- Are all parts of SOW written so there is no question as to what the contractor is obligated to do, and when?
- When it is necessary to reference other documents, is the proper reference document described? cited? Is all of it really pertinent to the task, or should only portions be referenced?

C. PRELIMINARY PLANNING – CHECKLIST FOR SOW PREPARATION (CONT'D)

II

- Is there a time-phased data requirement for each deliverable item?
- Are proper quantities shown?
- Can SOW task/contract line items breakouts at lower levels be identified in sufficient detail so they can be summarized to discrete third-level WBS elements?

C. PRELIMINARY PLANNING – CHECKLIST FOR SOW PREPARATION (CONT'D)

II

- Have all requirements for data been specified
- Are security requirements adequately covered if required?
- Has its availability to contractors been specified?

C. PRELIMINARY PLANNING INFORMATION REQUIREMENTS

- The following core documents are considered:
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C. PRELIMINARY PLANNING – PROJECT SPECIFICATIONS

- A specification list can be identified separately or as part of the SOW
- Specifications are used for pricing out a project
- Specifications are used for man-hours, equipment and material estimates
- Specifications can be written in different ways: Open vs. closed specifications, performance vs. prescriptive specifications, etc.

C. PRELIMINARY PLANNING – PROJECT SPECIFICATIONS LIST

Descriptions	Specification No.
Civil	100 (Index)
• Concrete	101
• Field equipment	102
• Roofing and siding	121
• Soil testing	122
• Structural design	124
Electrical	200 (Index)
• Electrical testing	201
• Heat tracing	201
• Motors	209
• Power systems	225
• Switchgear	226
• Synchronous	227
HVAC	300 (Index)
• Hazardous environment	301
• Insulation	302
• Refrigeration piping	318
• Sheetmetal ductwork	319
Installation	400 (Index)
• Conveyors and chutes	401
• Fired heaters and boilers	402
• Heat exchanger	403
• Reactors	414
• Towers	415
• Vessels	416

Descriptions	Specification No.
Instruments	500 (Index)
• Alarm systems	501
• Control valves	502
• Flow instruments	503
• Level gauges	536
• Pressure instruments	537
• Temperature instruments	538
Mechanical	600 (Index)
• Centrifugal pumps	601
• Compressors	602
• High-speed gears	603
• Material handling equipment	640
• Mechanical agitators	641
• Steam turbines	642
Piping	700 (Index)
• Expansion joints	701
• Field pressure testing	702
• Installation of piping	703
• Pipe fabrication specs	749
• Pipe supports	750
• Steam tracing	751

C. PRELIMINARY PLANNING INFORMATION REQUIREMENTS

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C. PRELIMINARY PLANNING – MILESTONE SCHEDULE / ACTION PLAN

- Milestone Schedules contain such information as:
 - Project start date
 - Project end date
 - Other major milestones, e.g, review meetings, prototype available, procurement, and testing.

C. PRELIMINARY PLANNING – MILESTONE SCHEDULE

MEMO	
To allow Ajax to operate like a department of Instat by April 1, 1996, we must do the following by the dates indicated.	
September 24 Ajax management to be advised of coming under Instat operation. The Instat sales department will begin selling Ajax Consumer Division production effective Jan. 1, 1996. There will be two sales groups: (1) Instat, (2) Ajax Builder Group.	November 5 Instat regional managers of Ajax for sales training session. November 26 Walters visits Ajax to obtain more information
October 15 Instat Regional Managers advised- Instat sales department to assume sales responsibility for Ajax products to distribution channels, Jan. 1, 1986. October 15 Ajax regional managers advised of sales changes effective Jan. 1, 1996	November 30 Data Processing (Morrie Reddish) and Mfg. Engineering (Sam Newfield): Request DP tapes from Bob Cawley. Ajax, for conversion of Ajax to Instat eng. records: master inventory file, cross-file, bill of materials file, where-used file, cross-reference Instat to Ajax part numbers, etc. Allow maximum two weeks until December 14, 1995, for tapes to be at Instat
October 15 Instat Management, Bob Carl, Van Baker, and Val Walters visit Ajax management and plant. Discuss how operations will merge into Instat.	December 3 ADMINISTRATIVE (Val Walters): Offer Norwood warehouse for sublease.
October 22 Ajax regional managers advised Ajax sales personnel and agents of change effective Jan. 1, 1996. October 24 Brent Sharp and ken Roadway visit Instat to coordinate changeover. October 29 Instat regional managers begin interviewing Ajax sale personnel for possible positions in Instat's sales organization	December 3 SALES (Abbott and Crutchfield): Week of sales meeting... instruction of salespeople in Ajax line... including procedure in writing Ajax orders on separate forms from Instat orders... temporarily, adding wright and shipping information, and procedure below: Crutchfield to write procedure regarding transmission of orders to Instat, credit check, and transmission of order information to shipping point, whether Norwood, San Francisco, or later, Instat Cincinnati

C. PRELIMINARY PLANNING – MILESTONE SCHEDULE (CONT'D)

Objective: Merger of Ajax Hardware into Instat Corp. by April 1, 1996			
Steps	Due Date	Responsibility	Precedent
1. Ajax management advised of changes	September 24	Bob Carl, Van Baker	-
2. Begin preparing Instat sales dept. to sell Ajax Consumer Division products effective 1/1/96	September 24	Bob Carl	1
3. Prepare to create two sales groups: (1) Instat, (2) Ajax Builder Group effective 1/1/96	September 24	Bob Carl	1
4. Advise Instat regional managers of sales division changes	October 15	Bob Carl	2,3
5. Advise Ajax regional managers of sales division changes	October 15	Van Baker	2,3
6. Visit Ajax management and plan to discuss merger of operations	October 15	Bob Carl, Van Baker, Val Walters	4, 5
7. Advise Ajax sales personnel and agents	October 22	Van Baker	6
8. Visit Instat to coordinate changeover	October 24	Brent Sharp, Ken Roadway	6
9. Interview Ajax sales personnel for possible positions	October 29	Instat regional managers	7
10. Sales training sessions for Ajax products	November 5	Instat regional managers	9
11. Visit Ajax again	November 26	Val Walters	8, 10
12. Request DP tapes from Bob Cawley for conversion	November 30	Morrie Reddish, Sam Newman	6
13. Offer Norwood warehouse for sublease	December 3	Val Walters	11
14. Write order procedures	December 3	Doug Crutchfield	10
15. Sales meeting (Instruction-product line and procedures)	December 3	Fred Abbott, Doug Crutchfield	14
16. DP tapes due for master inventory file, bill of material, structure file	December 14	Bob Cawley	12
...			

II

C. PRELIMINARY PLANNING INFORMATION REQUIREMENTS

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- **Detailed Planning**
- Execution
- Testing and Commissioning

D. DETAILED PLANNING

- Involves the development of detailed project schedule and budget
 - Development of a Work Breakdown Structure (WBS)
 - Networking activities in WBS
 - Generating Detailed schedule of activities in WBS
 - Developing detailed cost estimates of activities in WBS
- Detailed plan will serve as the basis for monitoring project progress and controlling deviations from plan.

PROJECT PLANNING LIFECYCLE PHASES (BY KERZNER)

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E. PROJECT EXECUTION

- Phase includes
 - Monitoring progress
 - Schedule updating
 - Cost and schedule control

PROJECT PLANNING LIFECYCLE PHASES (BY KERZNER)

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F. TESTING AND COMMISSIONING

- Closing out is an important phase in project life cycle with objective:
 - Bringing project closure to agreed upon contract requirements
 - Preparing transition to operational phase
 - Analyzing overall performance with regard to financial data, schedules and technical efforts
 - Closing project office and selling off all resources assigned to project

F. TESTING AND COMMISSIONING (CONT'D)

- Planning for project termination includes:
 - Transferring responsibility
 - Acceptance by user/sponsor
 - Releasing resources
 - Closing out work orders
 - Preparing for financial payments
 - Documenting results to reflect the as-built product
 - Completion of historic reports and post project analysis